

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Let's study operating systems!



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.

Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Can Do Statements:

- I can talk about the concept of operating system.
- I can describe the operating system parts.
- I can explain briefly the different types of operating systems.

Ask your partner:

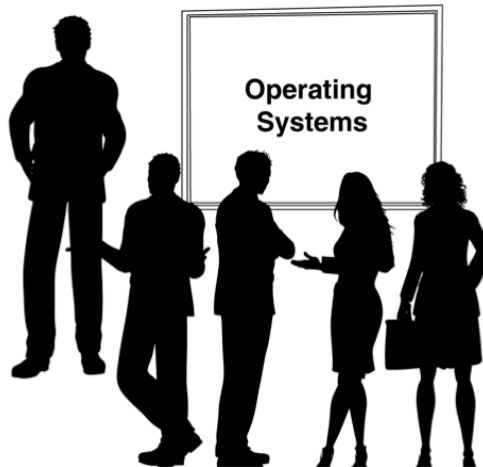
- Do you believe that the Operating System is the most important component of a computer? Why? Why not?
- Why do you think it is possible to work on different programs in a computer at the same time without them interfering with each other?





Let's Explore!

Professor Joseph on Operating System



Instructions: Read the text below.

Professor Joseph: Good morning, class.

Go to page 50 in your books, please. The Operating System or OS controls programs and applications in a computer. It performs many functions to run all the tasks inside a computer. This OS is just another program that can control automatically one task and transfer data or information to another. In other words, it manages the computer hardware.

At the beginning, programmers had to type the commands in order to perform a task. This took a lot of time and energy. Later, with the development of Operating System programs, the commands were easier to instruct. There are different types of OS like Batch Operating System, Time-sharing Operating Systems, Distributed Operating System, Network Operating System, and Real-Time Operating System.

Let me show you how it works. When you turn on your computer, the OS starts and controls all the applications you need to perform a certain task. For example, the control-card interpreter reads and carries all the commands when you click the "enter" key on the keyboard. Then, the loader will run all the applications and programs that you want to use. Finally, the drivers or device driver will check and operate all the input devices and output devices.

As you can see, the user interacts with the OS to instruct a command. For example, you want to rename or change the name of a document. In order to do that, you will introduce a command and this will be executed by the OS.

Ok, now, let's turn the page and take a look at the description of each of these Operating Systems.



**Let's Practice!**

Getting to Know the Main Types of OS

Instructions: Select the correct answer.

1. A type of OS is:
 - a. Analog Operating Systems
 - b. Time-sharing Operating Systems
 - c. Driver Operating System
2. An Operating System has the following function:
 - a. It control programs and applications in a computer.
 - b. It assists users in typing the commands to perform a task.
 - c. It turns on the computer.
3. What do the device drivers do?
 - a. They run all the applications you want to use.
 - b. They provide instruct commands.
 - c. They check all the input and output devices.

**Let's Practice!**

How an OS Works

*Instructions: Match the sentences halves a - d with their corresponding endings 1 - 4.
Read the sentences aloud. Check your pronunciation with your instructor.*

Let me show you how an OS works.

- a. When you turn on your computer, the OS starts and... ____
 - b. For example, the control-card interpreter reads and... ____
 - c. Then, the loader runs all the applications... ____
 - d. Finally, the driver or device driver checks and... ____
1. operates all the input and output devices.
 2. controls all the applications you need to perform a task.
 3. carries all the commands when you click the "enter" key on the keyboard.
 4. and programs that you want to use.





Extension Activity

Understanding What an OS Does

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find information about operating systems and their parts.
- Make a chart including what each part of the OS manages.
- Present your chart to the class.

Self-Assessment

Yes No

- I find information about operating systems and its parts.
- I understand what each part of the OS manages.
- I explain my ideas to the class.

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Target Words

- **point** (verb): to show or aim at something or someone in the horizon, on a screen, places, or a document
e.g. For example, the control-card interpreter reads and carries all the commands when you point or use the "enter" key on the keyboard.
- **rename** (verb): to change the name of a person, animal, or thing
e.g. For example, you want to rename or change the name of a document.
- **execute**(verb): to do or develop an idea, job, or order
e.g. you will introduce a command and this will be executed by the OS.
- **automatically** (adverb): when something occurs involuntarily, immediately, or very fast
e.g. This OS is just another program that can control automatically one task.
- **transfer** (verb): to move or take from one place to another
e.g. This OS is just another program that can control automatically one task and transfer data or information to another.



Learning Points

Simple Connectors

A **connector** is used to create connections between words and clauses. When you use them to join sentences, you give equal emphasis to these sentences you connect.

And, but, nor, yet, and **or** are simple **connectors** or conjunctions.

In this object, you are going to focus on the uses of **and**.

And is a **connector** or a linking word that joins two words, ideas, or phrases together.

Example:

I want a word processor **and** a spreadsheet in my computer.

He went to the store **and** bought a computer.

Examples

- *The Operating System was developed to control programs and applications in a computer.*
- *The control-card interpreter reads and carries all the commands when you point or use the "enter" key on the keyboard.*





Learning Points

Operating System: Types

The most commonly used types of operating systems are Batch Operating System, Time-sharing Operating System, Distributed Operating System, Network Operating System, and Real-Time Operating System.

- The Batch Operating System is an OS in which users create their job in an off-line device and which then submits the device to the computer. The problem is that it makes the computer slow.
- The Time-sharing Operating System is a way to work in which many people use the same software at the same time. In Time-Sharing Systems, the objective is to minimize response time.
- The Distributed Operating System uses multiple CPU to help multiple real-time applications and users. The different tasks are distributed among the processors accordingly.
- Network Operating Systems run on a server and offer the server the skill to manage data, users, groups, security, applications, and other networking functions. Examples of network operating systems include Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X, Novell NetWare, and BSD.
- The Real-Time Operating System is similar to a data processing system in which the time interval required to process and respond to inputs is so small that it controls the environment. There are two types, one called hard real-time systems which complete difficult tasks on time and the soft real-time systems which are less restrictive.